AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2016

Computer Science Paper II

Time: 2 hours 25 minutes Marks: 40

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

a. Define the term pseudocode. Write TWO advantages of using pseudocode before writing a computer program. (3 Marks)

b. Define the given types of computer error.

i. Syntax Error (1 Mark)

ii. Logical Error (1 Mark)

Q.2. (Total 5 Marks)

a. Write the value of the given GW BASIC expressions.

i. 8/4/2 (1 Mark)

ii. 8/(4/2) (1 Mark)

iii. 8*4/2 (1 Mark)

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b. State the function of the following statements of GW BASIC. (2 Marks)

Statement	Function
Left$	
On Error Goto	

Q.3. (Total 5 Marks)

Consider the given GW BASIC program.

```
10 LET SUM = 0
20 FOR A = 3 TO –3 STEP –2
30 SUM = SUM + A
40 NEXT A
50 PRINT SUM
```

a. Convert FOR NEXT loop into WHILE WEND loop in the given program. (4 Marks)

b. Also write the output of the program written in part a. (1 Mark)

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Q.4. (Total 4 Marks)

Consider the given marks of grade X students in Computer Science.

60, 55, 33, 22, 55, 70, 45, 59, 62, 58, 67, 57

Write syntax in GW BASIC to define:

i. A one-dimensional array (2 Marks)

ii. A two-dimensional array with 4 rows and 3 columns (2 Marks)

Q.5. (Total 3 Marks)

Write a user defined function in GW BASIC to calculate the value of $B^2 - 4AC$. Also print the value of the function for A = 4, B = – 5 and C = 1.

Q.6. (Total 3 Marks)

Using Boolean algebra theorems, prove that X + X.Y = X.

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Q.7. (Total 8 Marks)

EITHER

a. Suppose you connect a virus infected USB drive to your personal computer (PC).

i. Discuss at least SIX possible threats that your PC is exposed to.

ii. What steps should you take to avoid the viruses from infecting your PC?

OR

b. Artificial Intelligence (AI) is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the task of using computers to understand human intelligence.

Identify any FOUR fields of life where Artificial Intelligence (AI) is used. Also discuss ONE application of AI in each field.

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Q.8. (Total 7 Marks)

EITHER

a. Write a GW BASIC program to accept three numeric values from the user. Then find maximum and minimum values from them and print the range of these values.

Range = Max – Min

OR

b. Write a GW BASIC program to calculate and print average of three different numbers. The program takes input from the user. The program also determines and prints the quantity of entered numbers which are above and below the average.

Sample Output
A = 9, B = 27, C = 12

The average of three values: 16
The number of values above the average: 1
The number of values below the average: 2

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END OF PAPER